PAPER_721: UQFF Quantum Variable Documents: r_j, d_g, F_U, f_feedback, Omega_g

Class: `UQFFKnowledgeBaseKB6` **CP4 Entry:** #305 **Keywords:** UQFF, quantum variables, r_j, d_g, F_U, f_feedback, Omega_g, galactic spin, AGN feedback, magnetic string **Session:** 176 | **Version:** v5.33 **Source:** grok_share_ba508f76c8e.txt

Abstract

Five quantum variable document pages from the UQFF Knowledge Base define the foundational scalar quantities used across all UQFF master equations: r_j (magnetic string distance), d_g (Galactic Center distance), F_U (unified field strength total), $f_{feedback}$ (AGN feedback factor), and Ω_g (Galactic spin rate). Together they define the U_m and U_bi coupling to the Milky Way SMBH gravitational system.

1. Variable Registry

Variable	Value	Equation Role
r_j	1.496×10^{13} m (1 AU)	$U_m = \sum_j \frac{\mu_j}{r_j}(...)$
d_g	2.55×10^{20} m (\$\sim\$8 kpc)	$U_{bi} = -\beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g} ...$
F_U	2.28×10^{65} N	$F_U = \sum[k_i U_{gi} - \beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g} E_{react}] + ...$
$f_{feedback}$	0.1	$U_{g4} = k_4 \frac{\rho_{SCm} M_{bh}}{d_g} e^{-\alpha t} \cos(\pi t_n)(1 + f_{feedback})$
Ω_g	7.3×10^{-16} rad/s	$U_{bi} = -\beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g} ...$

2. Magnetic String U_m

$$U_m = \sum_j \left[\frac{\mu_j}{r_j}\left(1 - e^{-\gamma t \cos(\pi t_n)}\right)\hat{\phi}_j\right] \cdot P_{SCm} \cdot E_{react} \cdot (1 + 10^{13} f_H) \cdot (1 + f_{quasi})$$

3. Buoyancy-Gravity Coupling

$$U_{bi} = -\beta_i U_{gi} \Omega_g \frac{M_{bh}}{d_g}(1 + \varepsilon_{sw}\rho_{sw}) U_{UA} \cos(\pi t_n)$$

References

- UQFF KB grok_share_ba508f76c8e.txt entry #70
- Session 176, v5.33

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